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Multi-objective optimization of structural fire design

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ABSTRACT

Probabilistic risk assessment can be used for determining a design with a residual risk that is As Low As Reasonably Practicable (ALARP). Existing risk-based design approaches however predominantly focus on single objectives, which can be impractical considering the need for structures to meet diverse performance criteria across various dimensions, including monetary costs, environmental costs, and structural performance measures. To address these challenges, this study proposes the use of a multi-objective optimization (MOO) approach within a risk-based framework. Within the MOO framework, design goals for sustainability and resilience are incorporated together with safety objectives. The MOO approach is developed here for the risk-based design of structures exposed to fire. A multi-objective problem is formulated for a reinforced concrete slab in a multi-family dwelling by identifying design parameters, objectives, and constraints. Taking environmental costs into account has little effect on the optimum design obtained through MOO for the case study, except in cases where CO_2 emissions are highly valued. A limited increase in investment is observed to render the structure repairable after a fire incident, highlighting the importance of factoring in post-fire reparability in structural fire design.

1. Introduction

The structural design for fire exposure is traditionally determined through prescriptive design requirements [1]. However, with technological innovations and the continuous development of innovative structural materials, the use of prescriptive design methods may be inefficient in certain scenarios and conservative in others. This is because of our limited understanding and experience with innovative materials and design approaches. For these special designs, the adequacy should explicitly be demonstrated. In this regard, a performance-based design (PBD) regulatory system is recommended [2] which allows the proposal of customized design objectives. In the PBD approach, the design emphasis shifts towards attaining performance targets defined by stakeholders, rather than adhering strictly to prescriptive design criteria. Adopting probabilistic risk assessment (PRA) within PBD enables a thorough consideration of the uncertainties linked to both the structural design and its performance [3–7]. Furthermore, PRA explicitly allows for evaluating ALARP requirements and demonstrating design adequacy [8,9]. According to Ref. [9], the ALARP requirement entails a cost-benefit analysis, where safety investment cost is balanced with the benefits of risk reduction. The fundamental principle of this approach is that the resources conserved through cost-benefit analysis could be channeled into alternative tasks that would benefit society more (e.g., enhancing the healthcare system).

The PBD should present a comprehensive design on its own and, in the process of decision-making, ensure the fulfillment of all the relevant design objectives. Besides life safety [10], additional design objectives such as environmental protection, protection of infrastructure, and heritage values should also be taken into account during the decision-making [11]. Moreover, multiple stakeholders of the project could have their interests. For example, in the case of a building project, the owner may prioritize reduced lifetime cost, the contractor may emphasize minimized construction time, the occupants/renters could have an interest in lowering insurance premiums and minimizing maintenance-related downtimes, etc. The available studies that employ a cost optimization approach for structural fire design primarily focused on a single objective [12–15]. The fire design objectives in these studies are summed up to form a single optimization function. However, objectives cannot be directly added if they have different dimensions (monetary values, CO_2 emissions, etc.). Design objectives with a non-monetary dimension include the environmental cost commonly measured as CO_2 emissions, human losses measured in fatalities and injuries, repair downtime measured in the number of days of unavailability, etc. In these cases, the use of a single-optimization function is not feasible in the absence of commonly accepted conversion rates.

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List of abbreviations

ALARP	As Low As Reasonably Practicable
PBD	Performance-based design
PRA	Probabilistic Risk Assessment
MOO	Multi-objective optimization
SFE:	Structural fire engineering
SOO	Single-objective optimization
NSGA	Non-dominated sorting genetic algorithm
MAPE	Mean absolute percentage error
RC	Reinforced concrete
CDF	Cumulative density function

Furthermore, even if we succeed in developing a single objective function, several issues may arise when implementing single-objective optimization (SOO). The SOO can quickly become inefficient (requiring a large number of iterations) when large numbers of design parameters, objectives, and constraints are involved [16]. Furthermore, SOO gives a single solution and if there is a need to reconsider a conversion rate, the optimization needs to be carried out again. Importantly, because the objective function is often non-convex, SOO algorithms can lead to a local optimum [17]. A multi-objective optimization (MOO) can address these issues. MOO is a computational approach that facilitates decision-making for complex problems involving multiple conflicting objectives. A typical MOO problem has multiple design parameters, objectives, and constraints, see Ref. [17] and results to a range of Pareto-optimal solutions (set of optimum solutions). The objectives and constraints can have varied dimensions. The use of heuristic-based search algorithms can reduce the computational costs of optimization significantly [18,19]. Furthermore, MOO results in a set of solutions called Pareto-optimal solutions, and the optimum can quickly be evaluated if there is a change in specific preference of objectives.

MOO has been used in structural engineering for design optimizations. For example, MOO was applied to minimize investment costs and seismic risk in Ref. [20–22], for the maintenance and planning of reinforced concrete bridges in Ref. [23], and to minimize the environmental impacts of structural design in Ref. [24,25], and for planning and maintenance of construction projects in Ref. [26]. However, the application in the field of structural fire engineering (SFE) remains to be explored. The current study aims to establish a methodology for risk-based design optimization of fire-exposed structures that can consider multiple design parameters, objectives, and constraints simultaneously. In this regard, an MOO approach is proposed. To illustrate the approach, an application example of a reinforced concrete slab exposed to natural fire is considered. Additional fire safety design objectives targeting sustainability and resilience are considered for the MOO. Thus, in this study, MOO is demonstrated as a rational approach for evaluating optimum structural fire designs from multi-objective perspectives such as cost-effectiveness, sustainability, and resiliency.

2. Methodology of multi-objective optimization

MOO involves finding one or more optimum solutions in optimization problems with multiple objectives. MOO applies to problems with at least two conflicting objectives. Eq (1) presents a formulation for a general multi-objective optimization problem [17,27]. In the equation, $\boldsymbol{x} = [x_1, x_2 \ldots . x_n]^T$ represent a vector of n design variables. The x_i^L and x_i^U represent the upper and lower bound for the design variables and form the design space $\mathscr{D}$ of the MOO problem. The $g_j(\boldsymbol{x})$ and $h_k(\boldsymbol{x})$ are a set of inequality and equality constraints, and $f_m(\boldsymbol{x})$ represent a set of M design objectives which forms the objective space $\mathscr{Z}$.

$$\{Maximize \,/\, minimize\, f_m\,(\boldsymbol{x}), m = 1, 2, \ldots, M \text{ subjected to } \quad g_j\,(\boldsymbol{x}) \geq 0, j = 1, 2, \ldots, J; \quad h_k\,(\boldsymbol{x}) = 0, k = 1, 2, \ldots, K; \quad x_i^L \leq x_i \leq x_i^U, i = 1, 2, \ldots, n\} \tag{1}$$

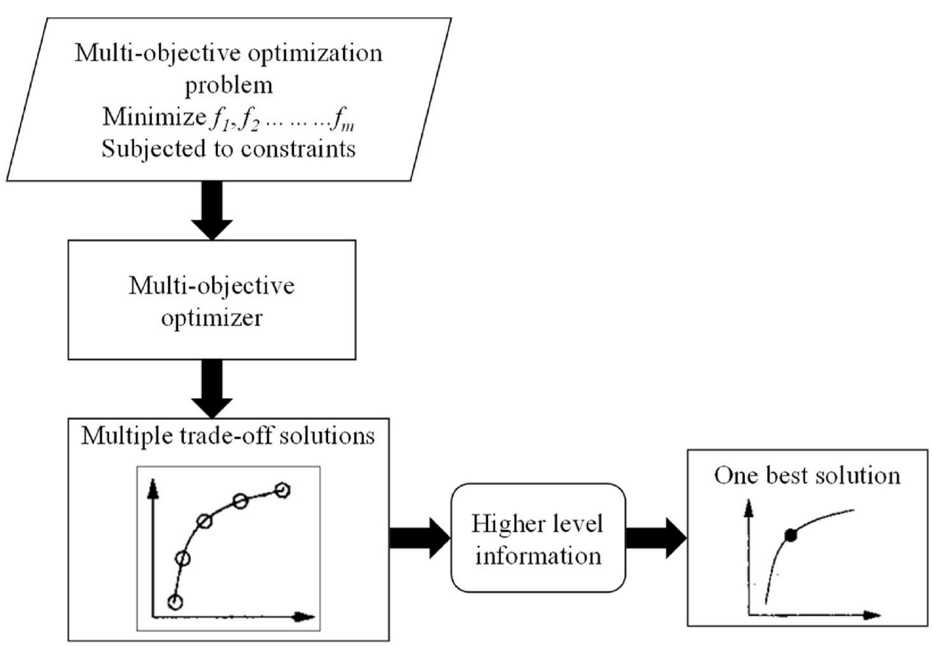

Fig. 1. A general flowchart for solving multi-objective optimization problems.

A general multi-objective optimization framework is presented in Fig. 1. Based on the approach, first, a multi-objective optimization problem is formulated by considering design parameters, objectives, and constraints. This is followed by adopting an optimization algorithm to find multiple trade-off solutions (also called Pareto-optimal solutions). The Pareto-optimal solutions are a non-dominated set of points, obtained through the principle of dominance. This principle entails that a non-dominated solution is no worse than any other solution for all the objectives. Lastly, the best solution is obtained by exploiting domain knowledge of the field, e.g., by using conversion rates between the objectives or by referring to direct stakeholder input for the trade-offs.

The existing MOO algorithms are commonly grouped into two types: (i) Classical and (ii) Evolutionary. In the classical approach, each of the objectives is assigned a preference (for example, weights) and finally combined to form a single composite objective function and it results in a single optimum solution. Thus, the classical approach is in reality a SOO with conversion rates translating the different objective dimensions into a single overall dimension for optimization. The evolutionary algorithm-based MOO initiates without requiring the specification of weights. They are inspired by nature's principles of evolution (such as natural selection) and use stochastic search techniques to obtain optimal solutions. Evolutionary algorithms result in multiple trade-off solutions. Commonly adopted evolutionary algorithms are genetic algorithms, simulated annealing, particle swarm optimization, etc. In this study, the non-dominated sorting genetic algorithm (NSGA-II) is adopted [21,22,28,29]. NSGA-II is chosen due to its ability to generate a diverse set of Pareto-optimal solutions, facilitating rapid evaluation of optimal solutions when varying design objectives. Unlike traditional genetic algorithms, NSGA-II maintains diversity in Pareto-optimal solutions through non-dominated sorting and crowding distance evaluation during offspring development in each generation of multi-objective optimization [17] While NSGA-II prioritizes multi-objective optimization, other evolutionary algorithms, such as particle swarm optimization and differential evolution, primarily focus on single-objective optimization.

3. Fire safety design objectives

Designing structures for fire relies on adhering to fire safety design objectives. Typically, stakeholders initiate a project by outlining their fire safety objectives. Mostly, they are qualitative (for example, type of building occupancy, downtime after a fire event, etc.) [30]. In general, the fire safety objectives specified in regulatory guidelines are regarded as sufficient for common design scenarios [31], leading to infrequent requirements for direct stakeholder consultations. However, in specific situations, it is crucial to incorporate the design objectives of all stakeholders, as regulatory guidelines may prove insufficient.

In [10] several fire safety objectives are outlined, including safeguarding occupants, ensuring structural integrity, and optimizing system effectiveness. These objectives primarily prioritize life safety in design, often neglecting other aspects like resilience and environmental protection, [11,30] provide a more comprehensive list of fire safety objectives. They include protection of health and life safety, protection of property, protection of the environment, protection of architectural, historic, or cultural values, and protection of infrastructure. To ensure the design's acceptability, it must showcase its performance across all these objectives in the event of a fire in the building. As a result, the comprehensive definition of fire safety encompasses these objectives. To ensure that project-specific societal fire safety objectives are not missed due to a tendency to refer to a limited set of commonly used predefined objectives, Van Coile et al. recommend considering an overarching objective of "reducing fire risk to socio-economically acceptable levels" [32]. This overarching goal is also listed in the *fib* state-of-the-art bulletin on performance-based fire design of concrete structures (*fib*, 2023). Specific fire safety objectives such as the preservation of cultural heritage and business continuity are then derived by considering the specifics of the project.

For demonstration purposes, two fire safety design objectives are considered for the MOO in this study. The first is the minimization of the lifetime cost, which is the sum of the investment cost for fire protection and the discounted cost resulting from fire consequences. The cost resulting from fire consequences takes into account the cost of damage to the structural system, non-structural system, building contents, and the cost of fatalities and injuries, as well as monetary indirect costs. The second objective is the minimization of environmental impacts of structural fire design. Taking into account these two design goals helps ensure adherence to the relevant design objectives mentioned in the previous paragraph for most common buildings. Besides, recently, there has been a growing emphasis on maintaining structural functionality following a fire incident, with a focus on enhancing structural resilience [33–37]. The post-fire reparability is therefore considered as a design constraint.

4. Case study: fire-exposed reinforced concrete slab

4.1. Case study description

A one-way loaded simply supported reinforced concrete (RC) slab is considered as a case study. The slab is from a compartment of 6 m × 6 m × 3.5 m of a multi-family residential dwelling. There are 3 stories in the building and each story has a floor area of about 5 times the size of the considered compartment. The slab is 0.2 m thick and is reinforced with 10 mm bars spaced at 100 mm center to center (i.e., area of reinforcing bars, $A_s = 0.000785$ m^2). The concrete cover to the reinforcement is 15 mm relative to the bottom face. The concrete has a characteristic strength of 30 MPa and reinforcement bars have a characteristic yield strength of 500 MPa. The same slab was considered in Ref. [38] for single-objective optimization under fire exposure.

4.2. Thermo-mechanical analysis

The capacity of the slab is evaluated as its resisting moment (M_R), based on Eq (2). In the equation, A_s is the area of reinforcing bars, a is the axis distance calculated as the distance of the center of reinforcement from the bottom face of the slab (i.e., $a = 15 + 10/2 = 20$ mm), and b is the slab width. The retention factor for yield strength of reinforcement is denoted as $k_{f_y,T}$, and $f_{c,20}$ and $f_{y,20}$ are the characteristic concrete strength and reinforcement yield strength at 20 °C, respectively. Eq (2) allows to evaluate the bending moment capacity of the slab both at ambient conditions and elevated temperatures. For ambient conditions, T is considered as 20 °C, while for fire exposure, a thermal analysis needs to be carried out. The same equation has also been adopted in Ref. [39,40] to evaluate the moment capacity of the slab.

$$M_R = A_s k_{f_y,T} f_{y,20}(h-a) - 0.5\frac{\left(A_s k_{f_y,T} f_{y,20}\right)^2}{b f_{c,20}} \tag{2}$$

At ambient conditions, the design moment capacity of the slab (M_{Rd}) is evaluated as 59 kNm (adopting 1.15 and 1.5 as safety factors for steel and concrete, respectively). A one-dimensional thermal analysis is carried out to determine the temperature of reinforcing bars during fire exposure. The entire cross-section of the slab is modeled as concrete and the material properties are obtained from Eurocode [41] for the thermo-mechanical analysis. The validation of the model to determine the temperature of reinforcing bars can be found in Ref. [42]. The fire exposure to the slab is based on the Eurocode parametric fire curve. The thermal inertia for walls is considered as 1450 J/(m^2 s$^{1/2}$ K), the same as in Ref. [40]. The temperature of the reinforcing bars is evaluated considering the burnout capacity of the slab [43], i.e., the maximum temperature (T_{max}) throughout the heating and cooling phase is evaluated as this results in the minimum capacity considering Eq (2).

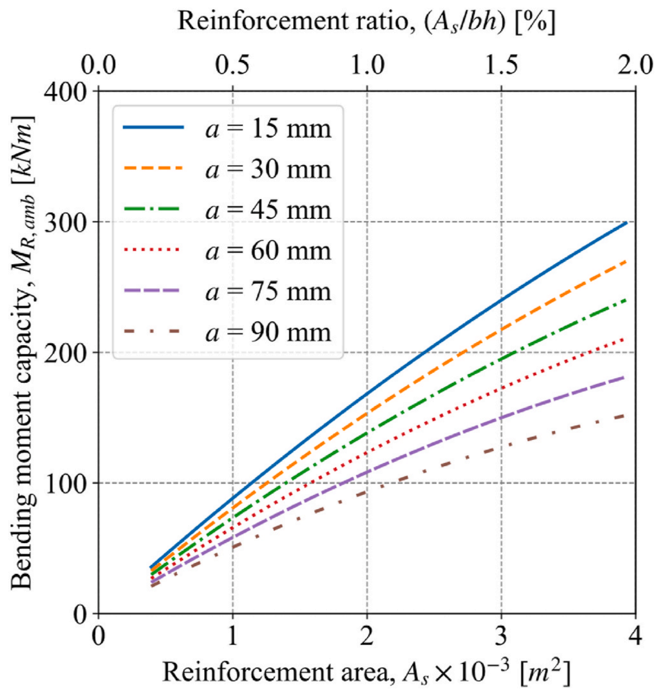

Fig. 2. Influence of design parameters (A_s and a) on the design bending moment capacity of the slab at ambient conditions. (b is slab width and h is its thickness).

5. Multi-objective problem formulation

5.1. Design parameters

Enhancing the fire resistance of a concrete structure is commonly achieved by increasing the axis distances (a) of the reinforcing bars, see Ref. [41] where specific recommendations for a to achieve a prescribed fire rating are listed. An increase in a reduces the temperature of the reinforcing bars during fire exposure, thereby limiting the reduction in the moment capacity for a simply supported slab. Another parameter enhancing the fire performance is the area of reinforcing bars, see Eq (2). A higher A_s results in a higher bending moment capacity and therefore, the slab has a higher fire resistance. Thus, a and A_s are considered as two design parameters for multi-objective optimization. The influence of the other design parameters on the bending moment capacity of the slab is relatively smaller.

The influence of a and A_s on the ambient design moment capacity of the slab at ambient temperature is presented in Fig. 2 (considering safety factors for steel and concrete as 1.15 and 1.5, respectively). The moment capacity increases with A_s, whereas it decreases for a higher a. This is because an increase in a reduces the moment lever arm in a slab with fixed depth and therefore reduces the moment capacity. Moreover, in Fig. 2, the moment capacity of the slab is found to be greatly affected by A_s, and as A_s increases, the influence of a becomes more prominent.

The influence of a on the minimum moment capacity of the slab during fire exposure can be deduced from Fig. 3 where parametric fires with varying q_f (fire load density) and O (opening factor) are considered as fire scenarios. In the figure, A_s is adopted as 0.000785 m^2 (reference case) and the partial safety factors for both concrete and steel materials as 1.0 based on [41]. The discontinuity in the calculated temperature and the moment capacity in the figure at lower temperatures is due to the change in fire from fuel-controlled to ventilation-controlled. In Fig. 3, the maximum temperature of the reinforcing bars is significantly reduced with an increase in cover from 20 to 60 mm. The slab's moment capacity ($M_{R,fi}$) is significantly higher for a of 20 mm than 60 mm for fires with a lower q_f, while the difference gets smaller with an increased q_f up to the point where the capacity is larger for $a = 60$ mm when the fire load density is very high. This is because, for fires with lower q_f, the effect of the smaller lever arm is more dominant than the effect of heating the reinforcing bars. For the MOO, the range for a is considered between 12 and 80 mm, and for A_s between 0.000393 m^2/m (0.2 %) and 0.00393 m^2/m (2 %). For the maximum considered reinforcement area (i.e., 0.00393 m^2/m), the slab is still under-reinforced, avoiding the failure mode of concrete crushing. Eq (2) thus is applicable for the range of design parameters considered.

5.2. Design objectives

5.2.1. Minimization of lifetime cost

Economic optimization is one of the important tools for structural design decision-making. The design target reliability indices for normal design conditions in Ref. [44,45] are based on cost-optimization. The cost-optimization formulation by Rackwitz is broadly adopted [46,47]. In the approach, the lifetime cost is optimized which is the sum of the cost of safety investment and the costs associated with the risk of structural failure [48].

5.2.1.1. Investment cost, including obsolescence cost. For the considered RC slab, the investment cost is the cost of the increased reinforcement area of the slab (ΔA_s) over the minimum considered value of A_s (i.e., 0.000393 m^2). The cost of reinforcement is evaluated based on [49], a database for building and construction costs. The cost of reinforcement (C_s) is 2.89 \$ per kg based on the United States national average and thus 22,686 per m^3. The investment cost is independent of the axis distance (a) as the slab thickness remains fixed. The obsolescence value for the reinforcement is evaluated considering an obsolescence rate (ω) of 0.022/year [50]. Since the obsolescence cost is a future entity, it needs to be discounted before adding to the upfront cost of reinforcement. A discounting rate (γ) of 0.03/year is adopted [51], and an infinite lifetime is considered for simplicity (i.e., assuming rebuilding after failure). Thus, the investment cost (C_I) for the RC slab (per m^2) can be calculated as:

$$C_I = C_s \times \Delta A_s \times (1 + \omega / \gamma) \tag{3}$$

For a unit m^2 of the slab, the cost for the maximum considered A_s (0.00393 m^2) is evaluated as 22,686 × (0.00393–0.000393) × (1 + 0.022/0.03) = 139 \$.

5.2.1.2. Evaluation of structural failure probability. The evaluation of the structural failure cost due to a fire event involves determining the structural failure probability ($P_{f,fi}$). The failure probability for the RC slab is calculated by considering flexural failure as the dominant failure mode of the slab. The limit state equation considering flexural failure is given by:

$$Z = K_R \times M_{R,fi} - K_E(M_G + M_Q) \tag{4}$$

where $M_{R,fi}$ refers to the slab's moment capacity for fire exposure and M_G and M_Q are the permanent and imposed load moments, respectively. K_R and K_E are the resistance and load effects' model uncertainties.

The probabilistic evaluation of the slab's moment capacity for fire exposure is based on Eq (2). While the calculation of $M_{R,fi}$ based on Eq (2) is relatively straightforward in terms of computational cost, the thermal analysis required to determine $T_{\max}$ is resource-intensive. The computational cost for probabilistic analysis increases significantly when multiple design parameters and fire scenarios are involved. Consequently, a surrogate model based on three-dimensional linear interpolation is developed where q_f, O, and a are input parameters and $T_{\max}$ is the output. The interpolation function is created by taking into account a range of values: q_f varies from 25 to 6025 MJ/m^2 with increments of 25 MJ/m^2, O spans from 0.02 to 0.2 m$^{1/2}$ with increments of 0.01 m$^{1/2}$, and a ranges from 1 to 101 mm with 1 mm increments. The

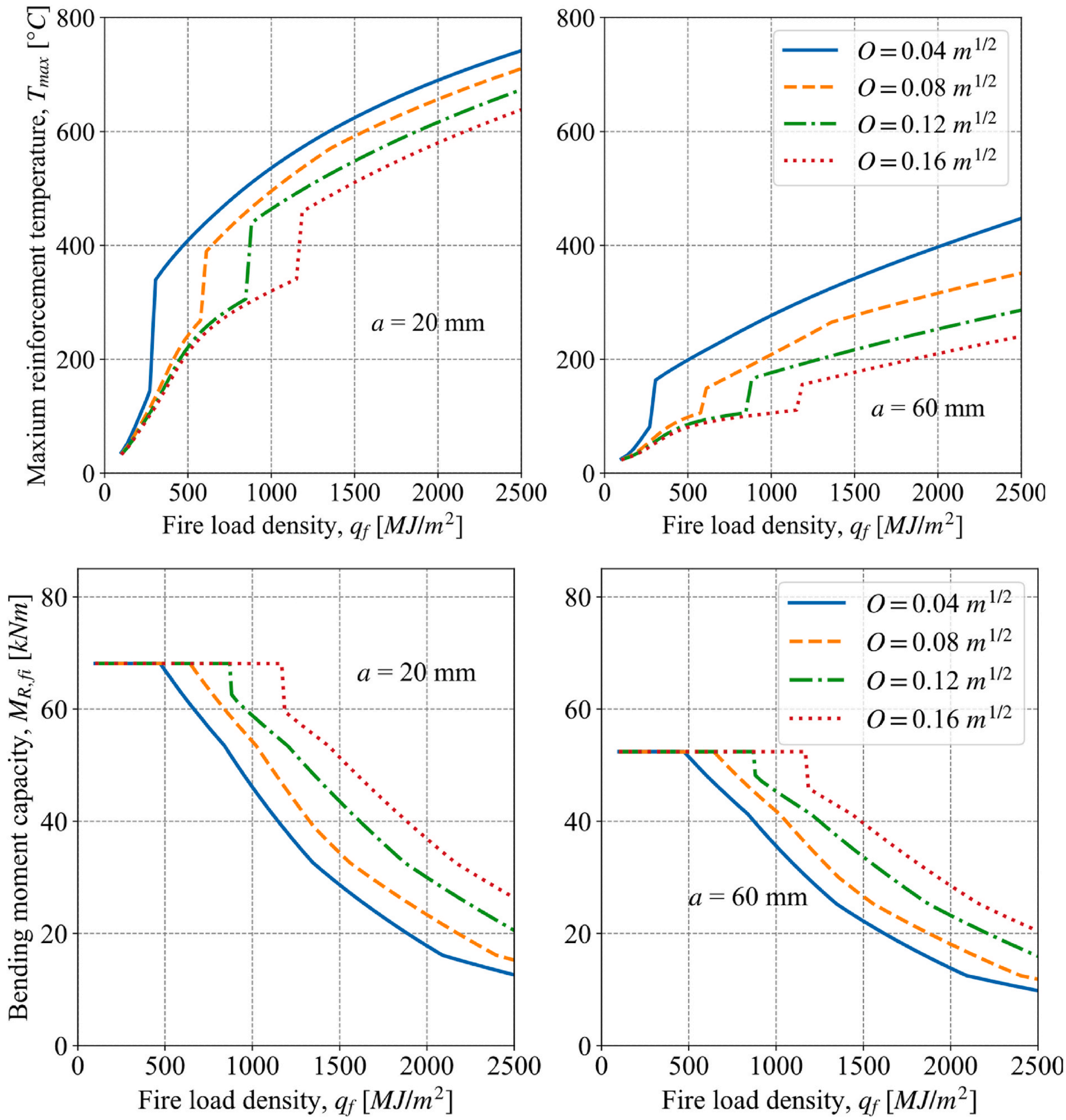

Fig. 3. Influence of a on the maximum temperature of reinforcing bars and the minimum bending moment capacity of the RC slab under parametric fire exposure. A_s is considered as 0.000785 m^2 (reference case).

interpolation function has a mean absolute percentage error (MAPE) of 1.6 %, with the errors mostly occurring in regions with fire scenarios changing from fuel controlled to ventilation controlled (with a maximum error of 60 °C). The error for other fire scenarios is almost zero. The interpolation function is formulated with the reference fire compartment dimensions of 10 m × 10 m × 3 m. When applying this function to any other equivalent compartment, it necessitates the use of a multiplicative conversion factor (ρ_{q_f}) for the fire load density. For compartments with the same thermal inertia as the reference compartment (as is the case here), this conversion factor is given by:

$$\rho_{q_f} = \frac{A_{f,eq}}{A_f} \times \frac{A_t}{A_{t,eq}} \tag{5}$$

where $A_{f,eq}$ and $A_{t,eq}$ correspond to the floor area and the total surface area of the specified (equivalent compartment), and A_f and A_t denote the floor area and total surface area of the reference fire compartment. The derivation for the conversion factor can be found in Ref. [40]. Table 1 lists the stochastic variables and their distribution for the probabilistic evaluation of $M_{R,fi}$. For the slab thickness, concrete strength, and reinforcement area, the probabilistic models proposed by Ref. [52] are adopted. a is modeled through a Beta distribution as suggested by Ref. [44], and q_f as a Gumbel distribution based on [53]. For the strength retention factor for yield strength of reinforcing bars, the model from Ref. [54] is considered. 10^5 realizations with a Monte Carlo (MC) approach are used for the probabilistic evaluation of $M_{R,fi}$.

The load moments on the slab are calculated by considering its full utilization in normal design conditions and by incorporating partial safety factors specified in Eurocode [53]. The characteristic moments from the permanent load (M_{Gk}) and the imposed load (M_{Qk}) are given by:

Table 1
Probabilistic parameters and their distribution for failure probability evaluation of the RC slab under fire exposure.

Variables	Symbol [unit]	Distribution	Mean, μ	SD, σ
Moment capacity, $M_{R,fi}$				
Slab thickness	h [m]	Normal	200	5
Concrete strength	$f_{c,\ 20^{\circ}C}$ [MPa]	Lognormal	42.9	6.4
Strength retention factor for reinforcement	$k_{fy,T}$ [−]	Logistic model [54]		
Reinforcement axis	a [mm]	Beta(4,4)	$a_{nom}+5$	5
Reinforcement area	A_s [mm^2/m]	Normal	1.02 $A_{s,nom}$	0.02 μ_{A_s}
Fire load density	q_f [MJ/m^2]	Gumbel	$q_{f,nom}$	0.3 μ_{q_f}
Opening factor	O [m$^{1/2}$]	Deterministic	0.04	–
Moment due to loads				
Permanent load	M_G [kN-m]	Normal	M_G	0.1 μ_{M_G}
Imposed load	M_Q [kN-m]	Gamma	0.2 × M_Q	0.95 μ_{M_Q}
Model uncertainties				
Capacity estimation	K_R [−]	Lognormal	1.1	0.11
Load estimation	K_E [−]	Lognormal	1	0.1

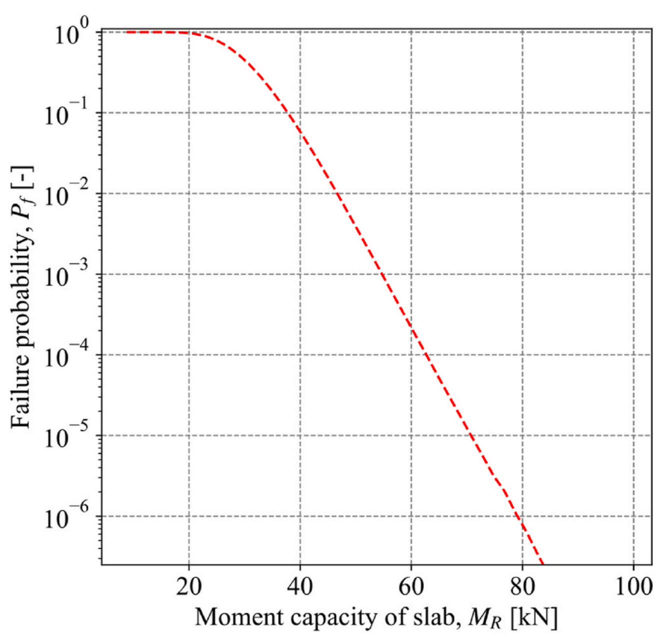

Fig. 4. Failure probability (1-F($M_{R,fi}$)) for the RC slab as a function of the moment capacity.

$$M_{Gk} = \frac{M_{Rd}}{max\left\{\left(\gamma_G + \psi_0 \gamma_Q \frac{\chi}{1-\chi}\right); \left(\xi_{rd}\gamma_G + \gamma_Q \frac{\chi}{1-\chi}\right)\right\}} \quad (6)$$

$$M_{Qk} = \frac{\chi}{1-\chi} M_{Gk} \quad (7)$$

where γ_G (with a value of 1.35) and γ_Q (with a value of 1.5) represent the partial safety factors assigned to permanent and imposed loads, respectively. The ψ_0 refers to the reduction factor for the imposed load action and has a value of 0.7 for the residential building. Similarly, ξ_{rd} is 0.85 corresponding to the reduction factor for the unfavorable permanent load. The load ratio (χ) is estimated as 0.30, assuming the permanent load on the slab as 4.8 (self-weight of the slab) + 1.6 (from other sources) kN/m^2, and the imposed load as 2 kN/m^2 (for residential buildings as recommended in Ref. [53]). The probabilistic modeling for the load moments considers Arbitrary-Point-In-Time load conditions as elaborated in Ref. [55], see Table 1.

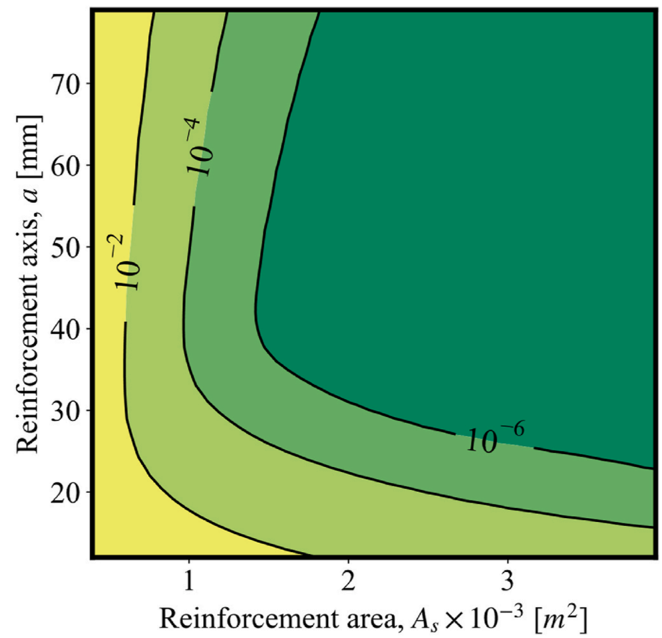

Fig. 5. Failure probability ($P_{f,fi}$) of the RC slab in function of the design parameters (A_s and a) for the reference fire exposure ($q_{f,\ nom}$ of 780 MJ/m^2 and O of 0.04 m$^{1/2}$).

For computational efficiency, the limit state equation for the RC slab represented by Eq (4) is modified to Eq (8). To evaluate the failure probability, a probabilistic distribution of Z_E is obtained by considering 10^8 MC realizations of K_E, K_R, M_G and M_Q. The failure probability for each realization of the $M_{R,fi}$ can be calculated based on Eq (9) where F_{Z_E} denote the cumulative density function (CDF) for Z_E. The relationship between failure probability and the slab's moment capacity is also presented in Fig. 4 from where $P_{f,fi}$ for each of the realizations of $M_{R,fi}$ can directly be interpolated. Lastly, the total failure probability is evaluated as the average $P_{f,fi}$ for all the 10^5 realizations of $M_{R,fi}$. In Fig. 4, the lowest failure probability considered is 2.5 × 10^{-7}, taking into account the 10^8 MC realizations for Z_E. This magnitude of $P_{f,fi}$ (2.5 × 10^{-7}) has a COV 0.2 for this number of simulations.

$$Z = M_{R,fi} - \frac{K_E}{K_R}(M_G + M_Q) = M_{R,fi} - Z_E \quad (8)$$

$$P_{f,fi} = P(M_{R,fi} - Z_E \leq 0) = 1 - F_{Z_E}(M_{R,fi}) \quad (9)$$

The failure probability for the RC slab as a function of a and A_s is shown in Fig. 5 for the reference fire exposure with nominal q_f of 780 MJ/m^2 and O of 0.04 m$^{1/2}$. Based on [56], 780 MJ/m^2 corresponds to the fire load density of residential occupancy. The $P_{f,fi}$ is significantly influenced by both a and A_s and has values ranging from 2.5 × 10^{-7} to 0.5.

5.2.1.3. Fire failure cost. The cost attributed to structural failure is modulated by the probability of failure given fire ($P_{f,fi}$) and the probability of fire occurrence (λ_{fi}) [38]. The lifetime failure cost due to structural fire exposure can be evaluated based on:

$$D = \frac{\lambda_{fi} P_{f,fi}}{\gamma} \xi C_{comp} \quad (10)$$

The failure cost represented by Eq (10) is the net present value (*NPV*) of the expected annual fire-induced failure cost considering an infinite lifetime for the structure. The same discounting rate (γ) as defined in Eq (3) is adopted. The likelihood of a fire event is evaluated as the rate of

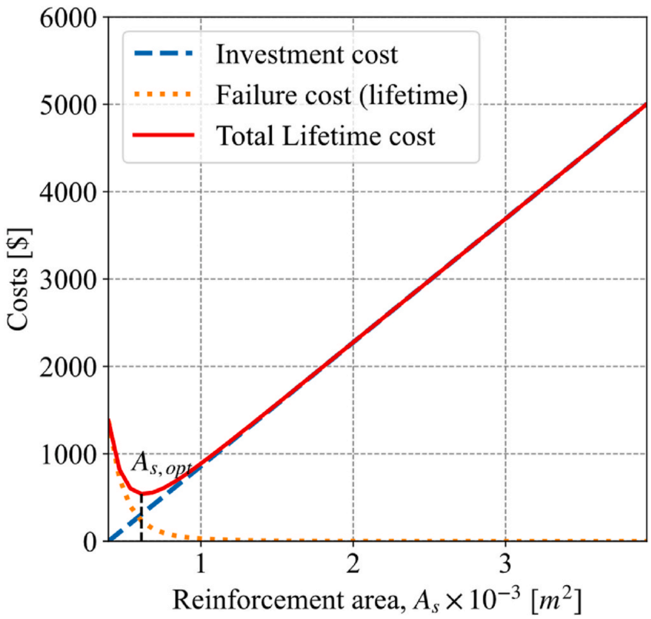

Fig. 6. Lifetime cost and its component in the function of reinforcement area of the slab for the reference fire exposure.

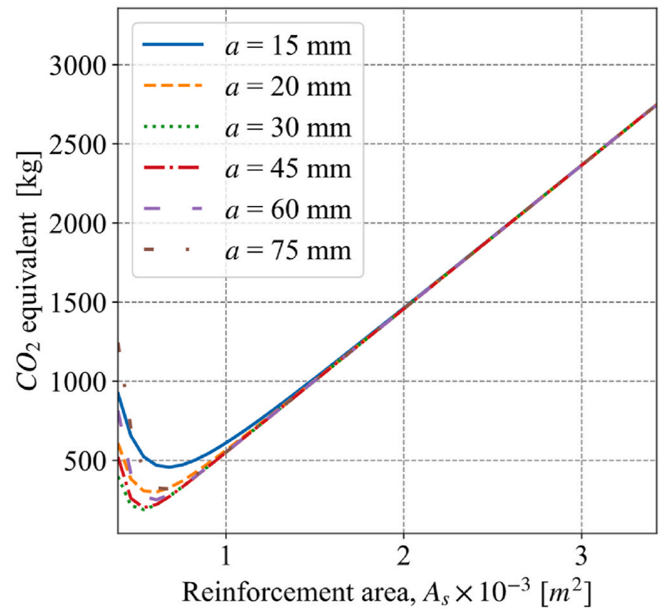

Fig. 7. Environmental cost as a function of design parameters (a and A_s) for the reference fire exposure ($q_{f,nom}$ = 780 MJ/m^2 and O = 0.04 m$^{1/2}$) of RC slab.

annual fire occurrence and a value of 0.00151/year is adopted based on [57]. For the reference compartment, λ_{fi} is calculated as 0.00151/15 (the building is 15 times the size of the compartment), i.e., 10^{-4}/year/compartment. Assuming absence of a sprinkler system in the building and fire and rescue service (FRS) located at a far distance and thus unable to prevent the fire from developing into a structurally significant fire, this value for λ_{fi} can be considered as the probability of structurally significant fires. Further in this study, the optimum designs are also evaluated for the building cases with the presence of a sprinkler system and FRS availability. When evaluating the λ_{fi} for the latter case, a failure probability for sprinkler systems (p^{spr}) and FRS is adopted as 0.11 [8] and 0.10 [58], respectively.

In Eq (10), ξ denotes the failure cost factor and is the ratio of the total cost following fire-induced structural failure to the construction cost of the compartment (C_{comp}). This factor includes the direct damage cost (structural system, non-structural system, contents, fatalities, and injuries) and the indirect costs. In the event of failure, the whole building is assumed to collapse and the direct damage cost for the structural system and non-structural system is thus the cost to replace the building (15 times the cost of the compartment). This assumption considers that the fire will spread upon structural failure, resulting in an even more challenging exposure for the remainder of the structure and thus a high likelihood of subsequent failures of other elements and compartments. Based on [49], the cost of construction of a multifamily residential dwelling (C_0) is 2314 \$/m^2 (215 \$/ft^2). The cost of the compartment is evaluated as $A_c \times C_0$ where A_c is compartment area. Additional direct damages relate to the cost of demolition and waste disposal, which is assumed as 5 % of the building cost, and the cost of contents amounting to 50 % of the direct structural and non-structural system costs and the disposal cost [59]. Fire-induced structural failure is considered not to significantly influence the risk to life for a low- or medium-rise building, acknowledging that the main risk to life is associated with toxic gasses. By varying the failure cost factor as part of a sensitivity analysis, the sensitivity of the results to such simplifying assumption can be explored. The indirect cost is challenging to determine. For the current study, the indirect cost is assumed equal to the direct damage cost, mainly attributed to the psychological damage as discussed by Ref. [60] for residential occupancies. Combining all the factors yields ξ as $15 \times 1.05 \times 1.5 \times 2 = 47.25$.

5.2.1.4. Lifetime cost. The lifetime cost (K_c) is evaluated as the sum of the investment cost and the structural failure cost due to fire events and can be calculated from Eq (11).

$$K_c = C_s \times \Delta A_s (1 + \omega / \gamma) \times A_c + \frac{\lambda_{fi} P_{f,fi}}{\gamma} \xi \times A_c \times C_0 \tag{11}$$

The lifetime cost and its component for the RC slab considering reference fire exposure are shown in Fig. 6 as a function of the reinforcement area. In the figure, the maximum investment cost is 5000 \$, while the failure cost is 1400 \$ and the optimum occurs for a reinforcement area of 606 mm^2. Once the reinforcement area exceeds about 1000 mm^2 no failure is expected to occur, however, this corresponds to an overinvestment. Note that the calculation is shown for a fixed axis distance (i.e., a of 20 mm, the reference case) and other values of axis distance can be associated with other optimum designs.

5.2.2. Minimization of environmental cost

The second design objective in the multi-objective optimization (MOO) is the minimization of the environmental impact of the structural fire design. CO_2 emissions are utilized as a cost indicator when quantifying environmental impact. For the considered RC slab, the environmental impact is evaluated as the sum of CO_2 from the increased reinforcement area (A_s) and from the risk of structural failure as represented by Eq (12). The equation has been formulated in accordance with Eq (11) and the evaluated impact can be termed as the lifetime environmental cost (K_e). A discounting factor γ_e is adopted for the NPV evaluation of the environmental impact of structural failure. Such discounting factor recognizes that climate change requires to limit emissions now. Thus an amount of emissions in 20 years is less environmentally detrimental than a nominally identical amount of emissions emitted today. This consideration results in Eqs. (11) and (12) appearing very similar. They have however different dimensions.

$$K_e = GWP_s \times \Delta A_s \times A_c + \frac{\lambda_{fi} P_{f,fi}}{\gamma_e} \xi_e \times A_c \times GWP_{tot} \tag{12}$$

In Eq (12), GWP_s denotes global warming potential of reinforcing bars of the slab (in CO_2 kg/kg) and GWP_{tot} denotes the global warming potential per unit floor area of the building (CO2 kg/m^2). ξ_e refers to the environmental failure cost factor and it is determined as the ratio of the floor area damage from the structural failure to the area of the considered

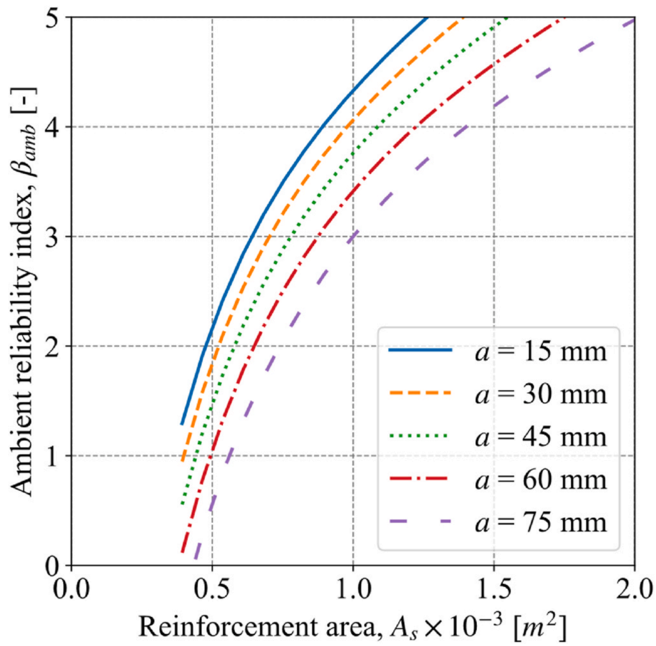

Fig. 8. Ambient reliability indices ($\beta_{t,amb}$) for the RC slab as a function of design parameters a and A_s. Evaluation considering a 50-year reference period.

compartment, similar to the failure cost factor. Based on [61], GWP_s per kg of steel equals to 3.21 $kgCO_2$ and GWP_{tot} is 168 $kgCO_2$ per m^2 of the floor area for the multi-family dwelling. The value for GWP_{tot} corresponds to the cradle-to-gate (product stage) CO_2 emission evaluation while in Eq (12), cradle-to-grave values are required (i.e., until the final disposal of the material). To account for this GWP_{tot} is increased by four times, corresponding to the GWP_{tot} for building materials for multi-family dwellings [61].

In the preceding section, the entire building is considered damaged in the event of structural failure and ξ_e therefore, equals to 15. A value of 0.014 is adopted as a discounting factor for the NPV evaluation of the environmental cost. Note that this value is lower than the discounting factor for monetary values. Kulczycka and Smol [62] discuss that monetary values and the ecological effects need to have a different discounting factor. However, no consensus exists on the discounting rate for environmental effects evaluation. Kula and Evans [63] and Stern [64] suggests a γ_e of 0.014. Fig. 7 plots the environmental cost for the fire exposure of the RC slab in the function of design parameters (a and A_s). The environmental cost on the one hand rises with an increase in A_s due to the direct expenditure of resources, while at the same time reducing with an increase in A_s due to the reduction in the risk of structural failure and the environmental costs associated with that failure scenario. Therefore, we can observe an optimum A_s in Fig. 7 for each considered axis distances. Furthermore, an a of 30 mm has the lowest environmental cost from all considered axis distances.

5.3. Design constraints

5.3.1. Ambient design

Optimization of a and A_s to enhance the fire resistance also influences the structural performance at ambient conditions. This study intends to evaluate the optimum design for fire exposure, while the ambient design requirements remain always fulfilled. Consequently, the ambient reliability index is used as a design constraint for the MOO. The value suggested by EN1990:2002 (target reliability i.e., $\beta_{t,amb}$ of 3.8) for a 50-year reference design period is adopted. When considering a 50-year design period, the imposed load is modeled through a Gumbel distribution (μ_{MQ} is 0.6 M_{Qk},COV of 0.35), see Ref. [55]. Note that the calculation of $P_{f,fi}$ takes into consideration the Arbitrary-Point-In-Time

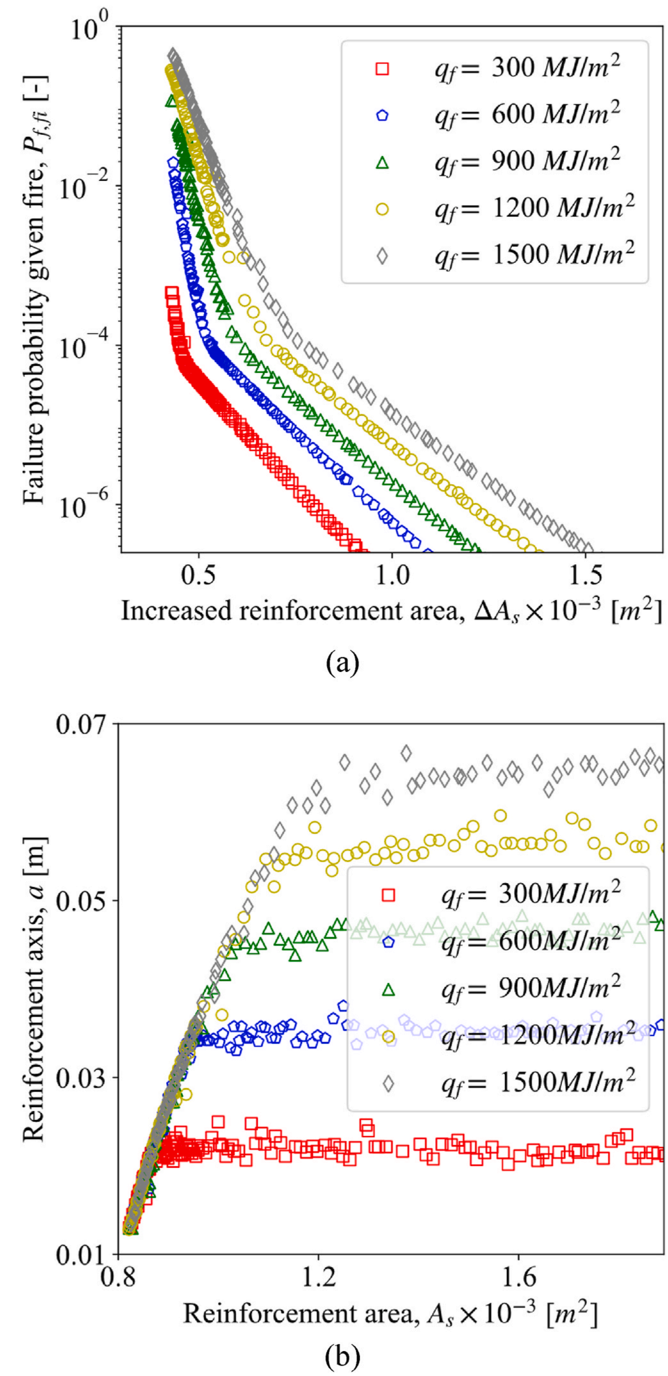

Fig. 9. (a) Pareto-optimal front and (b) solutions for the MOO of the RC slab, considering ΔA_s and $P_{f,fi}$ as design objective and $\beta_{t,amb}$ as design constraint.

load (see, Table 1) from Ref. [55]. The ambient reliability indices evaluated in the function of design parameters are illustrated in Fig. 8. The reliability indices are evaluated up to a maximum value of 5.0, attributed to the 10^8 realizations for the load as discussed earlier.

5.3.2. Post-fire repairability

Recently, there has been an increased emphasis on the need for rapid recovery and minimal loss of structural functionality following a fire [65,66]. Achieving this objective can involve incorporating post-fire repairability as a design constraint within the framework of multi-objective optimization (MOO). To apply this constraint, the maximum rebar temperature of 600 °C is considered as the repairability limit for the RC slab. This limit is based on the study by Ref. [67,68]

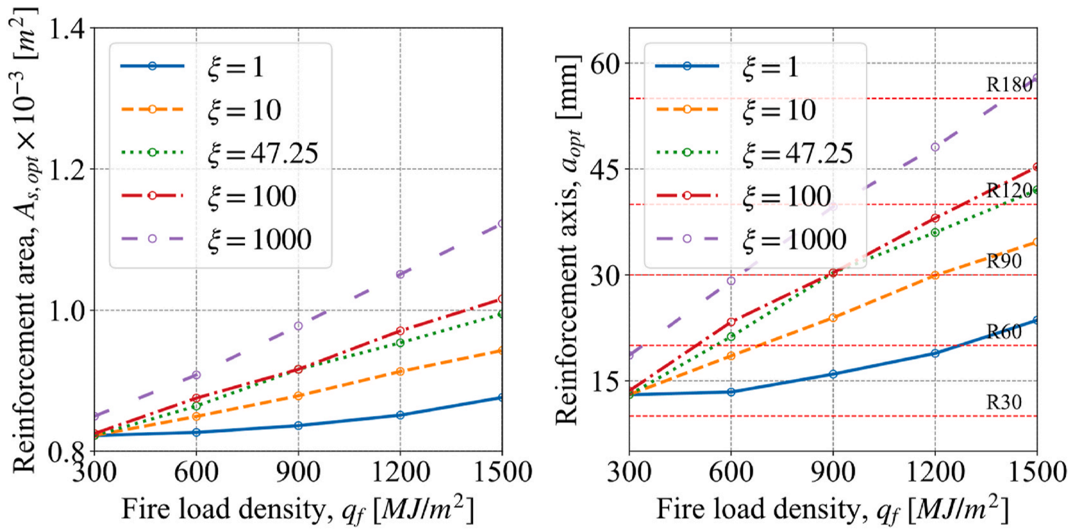

Fig. 10. Optimum A_s and a for the RC slab for different $q_{f,\ nom}$, and failure cost factors, ξ, without active fire suppression system ($\lambda_{fi} = 10^{-4}$/year/compartment).

which shows that reinforcing bars start losing strength permanently when exposed to temperatures higher than this limit. This criterion is implemented in the MOO framework through a probabilistic evaluation, considering the exceedance of 600 °C by the reinforcement's maximum temperature in the RC slab in case of fire as the limit state. The reliability index is evaluated for this limit state and compared with the limit state of serviceability (β_s) proposed for ambient condition in EN1990:2002 (a value of 2.9 is proposed for a 1-year reference period).

6. Multi-objective optimization

6.1. Implementation

As seen in Eqs (11) and (12), ΔA_s and $P_{f,fi}$ are the only variables dependent on the design parameters (ΔA_s, a). For computational efficiency, the MOO is thus implemented to minimize ΔA_s and $P_{f,fi}$ as intermediate objectives. The NSGA-II algorithm is then adopted to evaluate the pareto-optimal front and solutions. Once the Pareto-optimal fronts and solutions are obtained for the intermediate design objectives, they are substituted in Eqs (11) and (12) to calculate the Pareto-front for actual objectives (i.e., lifetime cost and environmental cost). As the other parameters of Eqs (11) and (12) are independent of the design parameters (ΔA_s, a), this step-wise procedure which has been adopted here for convenience and interpretability of the results does not affect the final outcome.

If all of the design objectives have the same dimensions (for example, monetary value for the investment cost and the failure cost), it is logical to consider the minimum of their sum as specifying the optimum solution. When the design objectives have dissimilar dimensions (e.g., environmental cost), a trade-off factor is necessary for combining design objectives to evaluate the optimum design. It is further highlighted that the objectives of Eqs (11) and (12) may appear very similar, but that their dimensions differ. Very different objective functions could be considered without modification to the methodology presented here.

6.2. Case 1: minimizing K_c as the design objective and $\beta_{t,amb}$ as design constraint

The lifetime cost K_c is minimized, considering the variables ΔA_s and $P_{f,fi}$ as intermediate design objectives. The Eurocode parametric fire with nominal $q_{f,\ nom}$ ranging from 300 to 1500 MJ/m^2 and a constant O of 0.04 m$^{1/2}$ is considered as fire scenario. The NSGA-II is applied as an algorithm for multi-objective optimization (MOO) with a population size of 150 and 30 offspring in each generation. The initial Pareto-

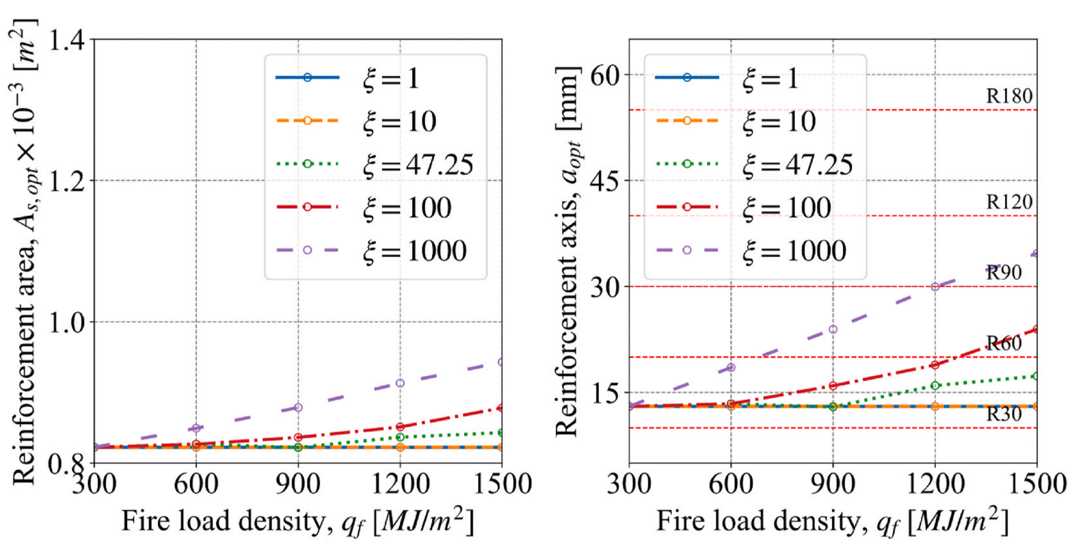

Fig. 11. Optimum A_s and a for the RC slab for different $q_{f,\ nom}$, and failure cost factors, ξ, with active fire suppression system ($\lambda_{fi} = 1.1 \times 10^{-6}$/year/compartment).

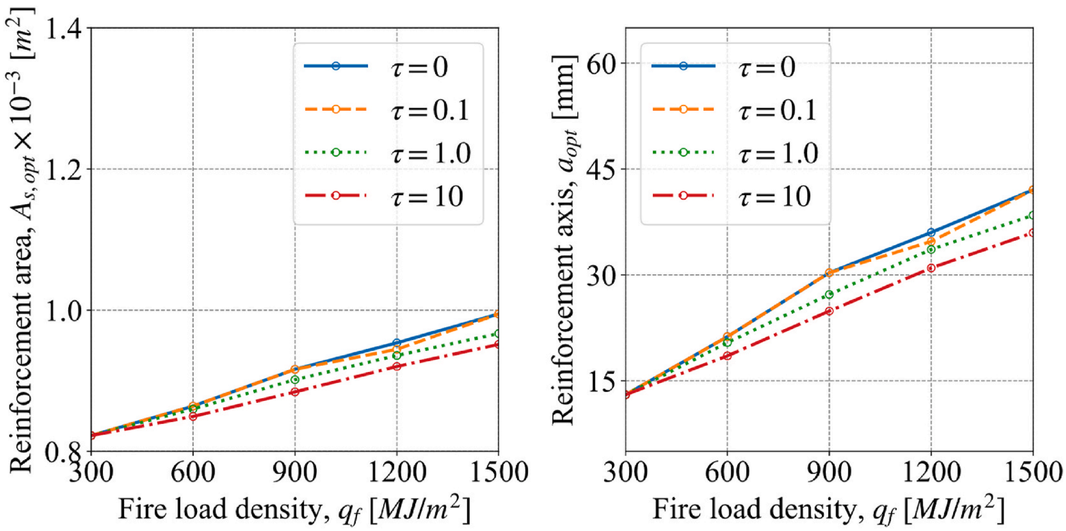

Fig. 12. Optimum A_s and a for the RC slab for different $q_{f,\ nom}$, and failure cost factors, ξ with the additional design objective of minimizing environmental cost (λ_{fi} 10^{-4}/year/compartment).

optimal solution does not have sufficient points in regions with smaller ΔA_s. To achieve a more diverse Pareto-optimal front, MOO is re-executed by considering a lower range of ΔA_s (0.000400 m^2–0.000500 m^2) and a population size of 50 with 20 offspring, and the resulting solutions are then merged. Fig. 9 presents the Pareto-optimal front and the solutions for the MOO of the RC slab. The solutions are the design parameters couples (ΔA_s, a) corresponding with minima of lifetime costs K_c, given a constraint on the reliability at ambient temperature $\beta_{t,amb}$.

As seen in Fig. 9(a), with an increase in $q_{f,\text{nom}}$, the optimum $P_{f,fi}$ increases significantly when ΔA_s is constant. An increase in investment (ΔA_s) reduces the optimum $P_{f,fi}$ to a considerably lower value for the considered fire scenarios. In Fig. 9(b), the corresponding value of a is visualized instead of $P_{f,fi}$. Interestingly a clear clustering of a can be observed with a change in $q_{f,\ nom}$. For low reinforcement area, the optimum axis distance is the same for all fire load densities. This optimum value increases with the reinforcement area, up to a point where the optimum axis distance remains approximately constant. This constant value is observed to be fire load dependent. This is because for low q_f the temperature ingress is limited, meaning that the reduction in temperature for higher a no longer compensates the reduction in lever arm.

Once the Pareto-optimal fronts for the intermediate objectives (ΔA_s and $P_{f,fi}$) are obtained, a normalized lifetime cost (K_c^*) can be evaluated based on Eq (13). The equation has been obtained by modifying Eq (11) where the constant term has been grouped into K_{comb}.

$$K_c^* = \Delta A_r + P_{f,fi} \times K_{comb} \tag{13}$$

where, $K_{comb} = \frac{\lambda_{fi} \times \xi \times C_0}{C_s \times \rho_s \times (\gamma + \omega)}$. With the Pareto-optimal front evaluated and using Eq (13), the optimum design can be evaluated for all the possible combinations of different parameters of K_{comb}. Herein, ξ is considerd a variable with values ranging from 1 to 1000 to account for uncertainties in the consequences of structural failure (47.25 as the reference case). For the scenario with the absence of an active fire protection system, λ_{fi} is adopted as 10^{-4}/year for the considered compartment, while λ_{fi} is evaluated as $10^{-4} \times 0.11$ (reduction factor for sprinkler system) $\times$ 0.1 (reduction factor for fire and rescue service) $= 1.11 \times 10^{-6}$/year for the case with the presence of an active fire suppression system, see Section 5.2.1. The values for the other parameters (C_0, C_s, ρ_s, γ, and ω) of K_{comb} were also discussed in Section 5.2.1.

The RC slab's optimal designs are shown in Fig. 10. This assessment is relevant to a specific building case in which there is no sprinkler system, and the fire-fighter service is located at a significant distance. Within the figure, for the reference case where ξ is 47.25 and q_f equals 780 MJ/m^2, the optimal values for A_s and a are 890 mm^2 (reinforcement ratio of 0.45 %) and 28 mm. With the consideration of the active fire suppression system ($\lambda_{fi} = 1.11 \times 10^{-6}$/year/compartment), for the reference RC slab, the optimum A_s and a are 820 mm^2 and 13 mm, respectively, as seen in Fig. 11 (as expected, lower values than the earlier case). In Fig. 10, $A_{s,opt}$ and a_{opt} are the same for $q_{f,nom}$ of 300 and 600 MJ/m^2 at ξ of 1.0 as the design parameter because these cases are governed by ambient design constraints ($\beta_{t,amb}$). For the case with the consideration of active fire suppression system (Fig. 11), the design parameters remain the same for all fire load densities for ξ equal to and below 10. For ξ of 47.25, $\beta_{t,amb}$ governs the design for $q_{f,nom}$ equal to and below 900 MJ/m^2. In these instances, the costs associated with failure are too small to justify offsetting the investment costs. In all the remaining scenarios, the design is driven by the optimization of lifetime costs.

Figs. 10 and 11 allow a comparison between the assessed MOO design and the tabulated fire resistance ratings listed in the Eurocode (EN1991-1-2:2002). Caution is advised since the Eurocode tabulated data should be considered to apply for the minimum reinforcement area required for ambient design. The comparison is nevertheless informative as it provides insight into the optimized nature of current guidance documents and legislation. As shown in the figures, an axis distance of 30 mm corresponds with a fire-resistance rating of 90 min according to the Eurocode tabulated data. In Fig. 10, this design is economical when ξ is equal to 1 (where damage equals the loss in the considered compartment) across all q_f values. However, if ξ exceeds 1, a greater value is necessary (i.e., a larger fire-rating should be required). For the reference case RC slab (when q_f equals 800 MJ/m^2 and ξ is 47.25), a 90 min of fire-rating is economical, see Fig. 10. However, when the presence of the active fire suppression system is considered, the fire rating requirement could be reduced to 60 min as seen in Fig. 11. Thus, the presence of an active fire suppression system could reduce the structural fire design requirement considerably. A 3-h fire rating of the RC slab is optimal when the consequence is equal to 1000 times the cost of the considered compartment (67 times the cost of the considered multi-family dwelling) and a $q_{f,\ nom}$ of about 1500 MJ/m^2. Thus, the Multi-Objective Optimization (MOO) approach can be utilized to assess the economic viability of current design provisions and suggest revisions. Furthermore, the design provisions relating to the structural fire design requirement considering different scenarios (presence/absence of active fire suppression system) can also be proposed.

Note that λ_{fi} in the case with consideration of active fire suppression

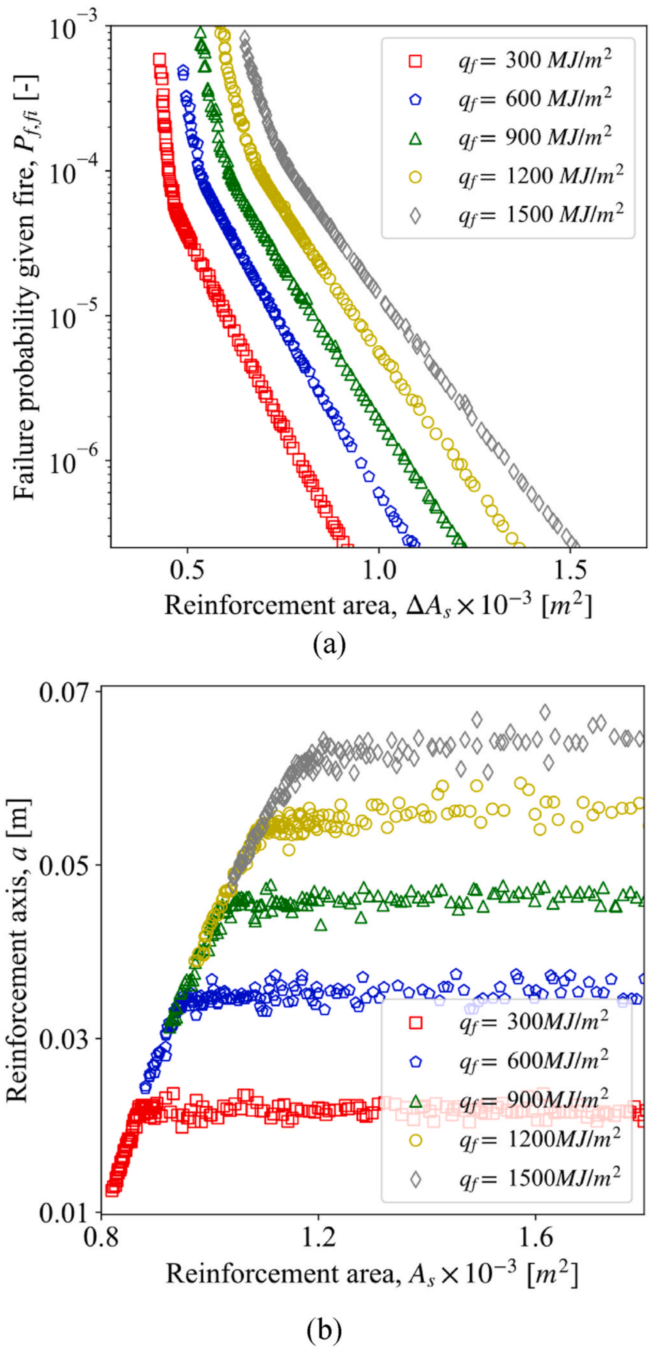

Fig. 13. Pareto-optimal front and (b) solutions for the MOO of the RC slab with ΔA_s and $P_{f,fi}$ as design objective and, $\beta_{t,amb}$ and post-fire repairability as design constraints.

system is about 100 times higher than the case with consideration of suppression system. Thus, the optimum design parameters for ξ of 1 in the former case are almost the same as for ξ of 100 in the latter case, see Eq (13). The result for the case with the consideration of an active fire protection system can thus easily be obtained by considering a corresponding increase/decrease in the failure cost factor from Fig. 10 as well. Consequently, in the next sections, the optimum design parameters are not evaluated for the cases with active fire suppression systems as they can be inferred from the cases without active suppression systems.

6.3. Case 2: minimizing K_c and K_e as design objectives and $\beta_{t,amb}$ as design constraint

To determine the overall optimum from the Pareto front solutions, we introduce a variable known as the trade-off factor (τ) to relate the CO_2 emissions with monetary values. This factor is a valuation constant to convert CO_2 emissions to dollars and has dimension \$ per kgCO_2. Determining τ is complex, as it relies on numerous variables like the country of origin and product source (local or imported), among others. The value of τ is assumed to range from 0.1 to 10. The Pareto-optimal fronts and solutions shown in Fig. 9 remain applicable for case 2 for MOO since ΔA_s and $P_{f,fi}$ are the intermediate objectives, see Eqs (11) and (12). The overall optimum solution is evaluated as the minimum value of $K_c + \tau \times K_e$ for each of the Pareto-optimal points. Detailed information on the parameters used to evaluate K_e can be found in Section 5.2.2. The optimum design parameter for the RC slab in function of fire scenarios and failure cost factor is displayed in Fig. 12. The failure cost factor (ξ) and environmental cost factor (ξ_e) are considered constant, with value of 47.25 (reference case) and 15 (damage is equal to the loss of entire building), respectively.

Fig. 12 illustrates that lower τ values (0 and 0.1) do not influence the optimal designs relative to Fig. 10. In such situations, the environmental cost does not affect the optimum design decision. In Fig. 12, higher τ values (1 and above) result in a decrease in the optimal values for the design parameters (A_s and a). This implies that the environmental cost is defined primarily by the CO_2 cost of placing additional reinforcement. The environmental cost due to structural failure is in other words small when contrasted with the CO_2 emission from the increase in the reinforcement area itself. The reason for structural failure cost being low is the rate of fire occurrence and the probability of structural failure being small. Thus, the consideration of the environmental cost in the MOO framework is concluded to result in a reduced safety investment. This reduction becomes significant when the trade-off factor is 1 \$ per kgCO_2 or higher.

6.4. Case 3: minimizing K_c as design objective, $\beta_{t,amb}$ and post-fire repairability as design constraints

The second constraint takes into account post-fire repairability for design optimization. This constraint is introduced in the MOO with β_s set at 2.9. Since the evaluation of Fig. 9 did not incorporate this design constraint, it is necessary to re-assess the Pareto-optimal fronts and solutions. Results are presented in Fig. 13. Like the previous set of Pareto-optimal solutions in Fig. 9, we can observe a substantial variation in A_s across the design space, while the Pareto optimal a steadily increases with the severity of the fire. Note that in the prior case (as depicted in Fig. 9), the Pareto-optimal values for a could be as low as 12 mm, even with a $q_{f,\,nom}$ of 1500 MJ/m^2. However, in the current case (Fig. 13), the minimum a is driven by the post-fire repairability constraint, which in turn is governed by the fire exposure scenario (q_f). As an example, for a $q_{f,\,nom}$ of 1500 MJ/m^2, the lowest a is about 50 mm.

Fig. 14 visualizes the optimum evaluated design parameters considering post-fire reparability as an additional design constraint. The A_s and a increase with the increase in fire severity ($q_{f,\,nom}$) in the figure, whereas ξ only has an effect for very high values above 100. In these instances, the governing factor for the optimal design is the post-fire repairability, denoted by β_s. The failure costs associated with these cases are much lower than the investment costs. For a significantly higher ξ (1000), the failure cost becomes significant in influencing design decisions, see Fig. 14 where there is an increase in the optimum design parameter values. In the reference RC slab (q_f = 780 MJ/m^2 and ξ of 47.25), the optimum A_s is 912 mm^2, indicating a 22 mm^2 (2.5 %) increase from the case when post-fire repairability constraint is not considered in the MOO. This implies that considering a design constraint of post-fire repairability could affect the investment in the slab at the design stage.

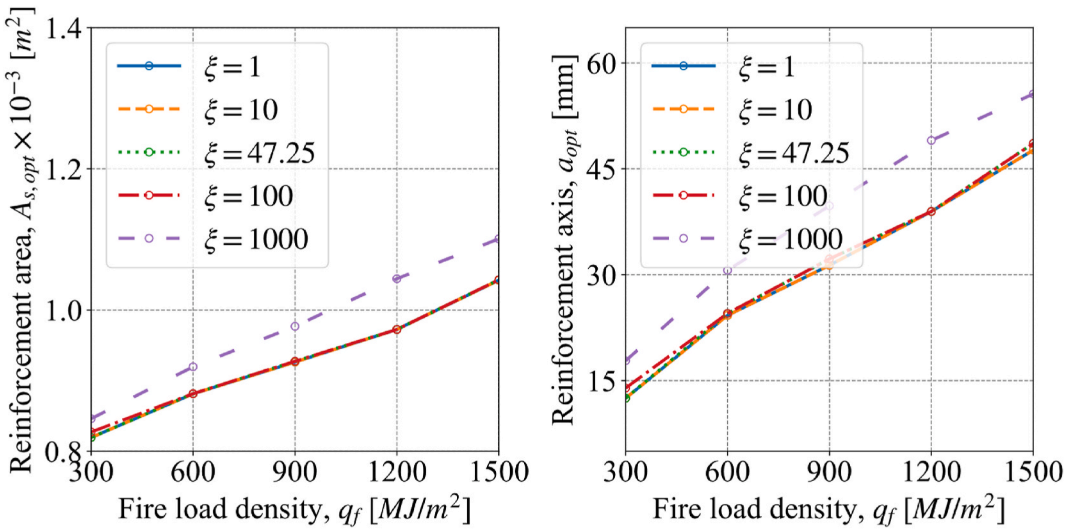

Fig. 14. Optimum A_s and a for the RC slab for different $q_{f,\ nom,}$ and failure cost factors, ξ with the additional design constraint of post-fire reparability (λ_{fi} 10^{-4}/year/compartment).

7. Conclusions

This study proposes a multi-objective optimization (MOO) approach for the risk-based design of structures exposed to fire. To illustrate the MOO, a case study of a reinforced concrete (RC) slab from a multi-family dwelling exposed to natural fire is considered. The multi-objective problem is formulated by identifying design parameters, objectives, and constraints, which address goals of safety, cost efficiency, sustainability, and resilience. Metrics for various fire safety design objectives, such as lifetime cost and environmental cost, as well as constraints such as ambient design target reliability and post-fire reparability, are proposed. The MOO approach is found flexible as the design objectives with dissimilar dimensions can be considered simultaneously, e.g., lifetime cost with monetary value and environmental cost with CO_2 emissions. Furthermore, the MOO enables evaluating the optimum design for any change in design objective parameters without a need for re-implementation.

For the slab, the optimum designs in terms of reinforcement area and axis distance were identified as a function of fire severity, failure consequences, design objectives, and design constraints. Sensitivity analyses on building occupancies and failure consequence factors including direct and indirect costs were considered. The environmental cost was considered through a trade-off factor relating the CO_2 emissions to monetary value.

For a concrete slab part of a multi-family dwelling, it was found that the incorporation of environmental cost did not influence the optimum structural fire design for values of the trade-off factor lower than 100 $ per ton of CO_2. When the environmental cost is increased to 1000 $ per ton of CO_2 or more, there is a decrease in the optimal design values. This decrease is because the environmental cost resulting from structural failure is smaller than the cost from increased material usage (here reinforcement). When considering post-fire repairability as a design constraint, this constraint dominates the design for the lower structural failure consequences. However, in situations with higher failure consequences, the design objectives decide the design. Interestingly, for the considered slab, a mere 2.5% increase in the initial investment cost could render it repairable after a fire event.

The current prescriptive design recommendations mainly rely on fire resistance ratings and limited distinction is made between occupancies. However, this study highlights the importance of customizing prescribed designs to align with both building occupancy and potential failure consequences. In the considered RC slab, a 20 mm (R60) reinforcement axis is cost-effective for all common building occupancies (nominal fire load density up to 1200 MJ/m^2) when ξ equals 1 (damage cost equal to the cost of the considered compartment). For a higher ξ, higher fire resistance ratings are economical. Incorporating an active fire suppression system in the design reduces the required fire rating generally by 30–60 min, notably for higher fire load densities. As a result, based on these findings, it becomes possible to update existing prescriptive design recommendations rationally. For generalization of the results, a sensitivity analysis is to be conducted by varying input parameters. Moreover, the results in the current study are presented purely from a cost-benefit perspective, while societal tolerability needs to be investigated as well.

CRediT authorship contribution statement

Ranjit Kumar Chaudhary: Conceptualization, Investigation, Methodology, Visualization, Writing – original draft. **Thomas Gernay:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Ruben Van Coile:** Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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